THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If the 4th term of an AP is 9 and the 9th term is 4, then the 13th term is: 1

(a) 0 **(b)** 1 **(c)** 36 **(d)** 13

2. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 2$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 1: 1

(a) 10 **(b)** 7 **(c)** 9 **(d)** 8

3. The point on the y-axis equidistant from $(2, 3)$ and $(6, 5)$ has $y =$ 1

(a) 12 **(b)** 14 **(c)** 13 **(d)** 11

4. The sum of the first 13 terms of an AP is 390, and its first term is 6. The 13th term is: 1

(a) 56 **(b)** 54 **(c)** 53 **(d)** 55

5. In a right triangle with hypotenuse $c = 5$ and one leg $a = 3$, the area (sq. units) is: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

6. The area of the triangle formed by $(7, 0)$, $(0, 9)$ and the origin, doubled, is: 1

(a) 64 **(b)** 63 **(c)** 62 **(d)** 65

7. In an AP, $S_8 = 200$ and $S_7 = 154$. Then $a_8 =$ 1

(a) 48 **(b)** 45 **(c)** 47 **(d)** 46

8. The altitude on the hypotenuse of the right triangle with legs 3 and 4 is of length (cm): 1

(a) $\frac{17}{5}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{7}{5}$ **(d)** $\frac{22}{5}$

9. If $P(x, y)$ is equidistant from $(1, 3)$ and $(7, 5)$ and lies on $y = 0$, then $x =$ 1

(a) $\frac{13}{3}$ **(b)** $\frac{22}{3}$ **(c)** $\frac{19}{3}$ **(d)** $\frac{16}{3}$

10. The sum of the first 10 terms of an AP is $2n^2 + 3n$. Its common difference is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

11. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 18$ and $ar(PQR) = 98$ and $AB = 6$, then $PQ =$ 1

(a) 16 **(b)** 15 **(c)** 13 **(d)** 14

12. The ratio in which $(-3, 2)$ and $(6, 5)$ are joined by the y-axis is 3: 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

13. The sum of the series $5 + 9 + 13 + \cdots + 101$ is: 1

(a) 1327 **(b)** 1326 **(c)** 1325 **(d)** 1324

14. In $\triangle ABC$ right-angled at B, $AB = 3$, $BC = 4$. If $BD \perp AC$, then AD (cm, correct to the exact value) is: 1

(a) $\frac{4}{5}$ **(b)** $\frac{19}{5}$ **(c)** $\frac{14}{5}$ **(d)** $\frac{9}{5}$

15. The side of the square with diagonal endpoints $(0, 0)$ and $(6, 8)$ has squared length: 1

(a) 50 **(b)** 49 **(c)** 52 **(d)** 51

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The 4th term of an AP is 9 and the 9th term is 4. Find the 13th term. 2

17. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $AB = 7$ cm and $BC = 14$ cm. Find DE. 2

18. Show that the points $A(0, 0)$, $B(3, 4)$, $C(8, 4)$, $D(5, 0)$ are the vertices of a rhombus. 2

19. If $S_n = 3n^2 + 4n$, find the first term and the common difference. 2

20. The areas of two similar triangles are 27 cm² and 75 cm². If a side of the first is 3 cm, find the corresponding side of the second. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In an AP, $S_6 = 75$ and $S_{10} = 185$. Find the AP. 3

22. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 8$ cm and $DC = 12$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

23. Show that the points $A(1, 1)$, $B(5, 4)$, $C(10, 4)$, $D(6, 1)$ are the vertices of a rhombus. 3

24. Find the sum of the first 16 terms of the AP whose 3rd term is 14 and 8th term is 34. 3

OR

Find the altitude of an equilateral triangle of side 6 cm, in terms of $\sqrt{3}$. 3

25. The line joining $(0, 0)$ and $(9, 6)$ is trisected by points P and Q. Find their coordinates. 3

OR

Find the sum of all integers between 10 and 200 that leave remainder 3 when divided by 7. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A ladder has 21 rungs whose lengths decrease uniformly from 100 cm at the bottom to 40 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

OR

Two ships leave a port at the same time; one sails 60 km due north and the other 80 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. 5

27. The vertices of $\triangle ABC$ are $A(0, 0)$, $B(12, 0)$ and $C(6, 8)$. Find the coordinates of the centroid and the area of the triangle. 5

OR

A loan is repaid in 18 monthly instalments; the first is ₹2500 and each later instalment is ₹150 more than the previous. Find the instalment in the last month and the total amount repaid. 5

END OF QUESTION PAPER — **Total: 50 Marks**